

# Aritra Mahanty

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## SKILLS

**Languages:** Python, C++

**Core Domains:** Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning

**Libraries:** PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, Gymnasium

**Tools:** Git, AWS, Linux, SQL

## EDUCATION

**Vellore Institute of Technology**

Bhopal, MP

*Bachelor of Technology – Computer Science and Engineering (AIML); CGPA: 8.43/10.0*

*Aug 2023 – May 2027*

## PROJECTS

**Hybrid Retrieval & Recommendation Engine**

Jan 2026 – Present

*Information Retrieval & Machine Learning*

*Python, SQLite, Gemini API, HNSWlib*

- **Data Pipeline:** Built an attribute extraction that used LLM and used  $k$ -NN to extend the LLM coverage over the less popular 52K plots, expanding semantic coverage from 15% to 80%.
- **Hybrid Retrieval:** Designed a 3-lane retrieval system (dense HNSW, tag-genome, BM25) fused via Reciprocal Rank Fusion for semantic, thematic, and keyword queries, boosting diversity by 141%.
- **Reranking:** Offloaded 150-candidate pool ranking from generative LLMs to a Reranker, eliminating context-window token overhead and position bias.

**Autonomous RL Agent for Prince of Persia**

Oct 2025 – Present

*Reinforcement Learning & Systems*

*Python, PyTorch, Gymnasium, ctypes*

- **C to Python Bridge:** Created the first Environment For PoP using ctypes to Bridge the Environment with SDLPoP sustaining 13K steps/sec across 27 async workers.
- **Agent Training:** Achieved 100% Level 1 completion (95% stochastic) using PPO and FiGAR variable-duration actions trained over 150M frames with a 12-channel observation grid.
- **Exploration System:** Engineered a progress dependednt reward system (milestone shaping, curiosity, memorized return paths) to solve multi-room navigation and backtracking across 24 rooms.

**Real-Time Facial Recognition Attendance System**

Jan 2025 – Mar 2025

*Computer Vision & Machine Learning*

*Python, OpenCV, ONNX, FAISS*

- **Pipeline:** Built a real-time facial recognition pipeline using YuNet detection with 5-point landmark alignment and SFace embeddings (14.7ms end-to-end latency).
- **Accuracy:** Validated 98.0% accuracy on LFW and 98.3% on Olivetti Faces via 5-fold CV, utilizing CLAHE illumination normalization on 128-dim embeddings.
- **Vector Search:** Indexed embeddings with FAISS L2, achieving  $8.6\mu\text{s}$  query latency and a  $3.7\times$  speedup over brute-force NumPy search.

## LEADERSHIP EXPERIENCE

**Tech Co-Lead – Bit by Bit VIT Bhopal**

Sep 2025 – Jun 2026

*Campus Technical Event Organization*

- **Technical Leadership:** Led a 10-member technical team as Tech Co-Lead, driving website development and technical infrastructure for club events, including a 200+ participant hackathon and expert talks.

## CERTIFICATIONS AND BADGES

**AWS Certified Cloud Practitioner (CLF-C02):** Amazon Web Services

Feb 2026

**Generative AI using IBM Watsonx:** IBM Career Education Program

Jun 2025

## ACCOMPLISHMENTS

- Top 10 Finalist – National Cyber Shield Hackathon, MP Police & ClearTrail (SBI-sponsored), Nov 2025
- Top 10 Finish – Hackcrux Hackathon, LNMIIT Jaipur, Mar 2025
- Merged Open-Source Contribution – JAX (Google's ML Framework), Bug Fix #31101 (PR #31192), Sep 2025